

DIGITAL MARKETING

PROJECT 2 — SEO BASICS & KEYWORD RESEARCH

Industrial Training Kit | DecodeLabs | Batch 2026

Search Intent, Long-Tail Strategy & a Verified Keyword List

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1. Project Overview

DecodeLabs' Project 2 moves an intern from social media 'push' tactics into search-driven 'pull' marketing. The brief is simple on paper — perform basic keyword research for a website or blog — but the underlying skill is Search Intent: understanding not just what people type into Google, but why they typed it. This report documents the full research process for a tech-internship and data-science training platform positioned in the same space as DecodeLabs itself, and delivers a verified shortlist of 20 target keywords ready to brief into content.

The site modeled throughout this report sits in the 2026 target domain defined in the training brief: programming languages (Python, SQL, Java), in-demand roles (Data Scientist, AI/ML Architect, Full Stack Developer), core skills (Pandas, NumPy, TensorFlow, React, Node.js), and common project queries (real-time chat apps, e-commerce stores). The competitive frame is explicitly David vs. Goliath — a focused niche player going after the specific technical queries that a generalist giant like Internshala is too broad to own.

2. Objective & Scope

Goal: identify a set of relevant, low-competition, high-intent keywords that a growing tech-education site can realistically rank for, and map each one to the type of content it should drive.

Key requirements covered in this report:

- Find relevant keywords across the 2026 target domain (languages, roles, skills, project types).
- Identify search intent for every keyword using the Informational / Navigational / Commercial / Transactional framework.
- Suggest keywords for content, each paired with a concrete content angle.
- Apply the Keyword Golden Ratio (KGR) to separate genuinely winnable keywords from ones that only look easy.

3. Understanding Search Intent

Every query a user submits is, functionally, a small program running through three stages — the same Input → Process → Output logic that governs search engines themselves:

Input: the user's query (demand)

Process: the algorithm's relevance + authority scoring

Output: the SERP (the result the user actually sees)

Ranking well means synchronizing content with that Input — which starts with correctly labeling what the searcher actually wants. Every keyword in this report is classified into one of four intent archetypes:

The Four Archetypes of Intent

Archetype	Searcher Is...	Example Query
Informational	Seeking knowledge	"What is an API?"
Navigational	Looking for a specific page	"DecodeLabs Login"
Commercial	Comparing options	"Best Python Course 2026"
Transactional	Ready to act	"Apply for Internship"

A single seed topic — say, "data science internship" — can spin off keywords in all four buckets: an informational "what does a data science internship involve", a navigational "DecodeLabs internship login", a commercial "best data science internship platforms for B.Tech students", and a transactional "apply for data science internship with stipend". Matching content format to intent — a guide vs. a comparison vs. a landing page — is what separates a page that ranks from one that ranks and converts.

4. Research Methodology

The research followed the five-step execution workflow below — the same sequence used by working SEO practitioners, scaled down to what's realistically checkable without a paid Keyword Planner / Ahrefs seat.

#	Step	What It Involved
1	Brainstorm	Seed keywords pulled from the target domain: languages, roles, skills, project types.
2	Expand	Variations generated the way Google Autocomplete does — pairing each seed with real modifiers (salary, roadmap, for freshers, remote, apply, free certificate, vs.).
3	Validate	Checked the live SERP for each candidate to see who currently ranks and how crowded the space is.
4	Calculate	Applied KGR = allintitle results ÷ monthly search volume to flag genuinely low-competition terms.
5	Categorize	Assigned one of the four intent archetypes to every keyword and paired it with a content format.

Tools referenced

- Google Keyword Planner — the source-of-truth for real monthly search volume (requires a Google Ads login).
- Ahrefs / Ubersuggest — competitor keyword analysis.
- Google Autocomplete — the ‘alphabet soup’ method for surfacing real long-tail phrasing.
- The allintitle: search operator — validates exactly how many pages are actively targeting a phrase in their title tag.

The Keyword Golden Ratio (KGR)

KGR = allintitle results ÷ monthly search volume. It is a fast proxy for “how contested is this exact phrase, relative to how many people search it.”

- KGR < 0.25 — Great. A new or small site can realistically rank within days to weeks.
- KGR 0.25 – 1.00 — Moderate. Winnable, but expect more competition and a longer timeline.
- KGR > 1.00 — Bad. Too competitive for a low-authority site to target directly.

Specificity is the lever that keeps KGR low: research cited in the training material notes that roughly 91% of all web searches are long-tail (three or more words), and long-tail queries consistently pair lower search volume with dramatically lower competition and higher conversion probability than short ‘head’ terms.

5. Target Domain Analysis

The 2026 target domain, as scoped in the training brief, breaks into four content pillars:

Category	Focus Areas
Languages	Python (23% developer share), SQL, Java
Roles	Data Scientist, AI/ML Architect, Full Stack Developer
Skills	Pandas, NumPy, TensorFlow, React, Node.js
Project Queries	Real-time chat app, e-commerce store

Competitive framing: David vs. Goliath

A generalist platform such as Internshala captures roughly 4.93M visits on the strength of broad terms and a generalist positioning. A niche player cannot out-broad that. The winning strategy is the inverse — target the specific technical terms and deep, well-structured content that a generalist site has no incentive to build out. Every keyword recommended in Section 6 is chosen with this asymmetry in mind: narrow enough to be winnable, specific enough to convert a genuinely qualified visitor.

6. Keyword Research Findings — 20 Verified Keywords

The table below is the core deliverable: 20 long-tail keywords spanning all four intent archetypes, each carrying an estimated search volume, a KGR-based competition verdict, and a concrete content angle. Volume and competition figures are research estimates from live SERP checks (see Section 10) — flagged for final confirmation in Keyword Planner before a content calendar is locked in.

#	Keyword	Intent	Est. Volume (India)	KGR Verdict	Suggested Content Angle
1	Data Scientist salary in India 2026 for freshers with Python and SQL	Commercial	200–500	Great	Fresher-specific salary breakdown, Python/SQL angle
2	Python vs Java for backend development for B.Tech students	Commercial	100–300	Great	Comparison guide framed around choosing a first backend language
3	Remote Data Science internship for freshers with stipend	Transactional	300–800	Moderate	Internship landing page, stipend & eligibility above the fold
4	Free Python internship certificate for engineering students	Transactional	200–500	Great	Landing page bundling free track with certificate preview
5	AI/ML Architect internship syllabus and skills required	Informational	50–150	Great	Syllabus page mapping skills to weekly milestones
6	How to become an AI/ML Architect after B.Tech	Informational	100–250	Great	Step-by-step roadmap post with internship CTA
7	Best remote internship platforms for B.Tech CSE students	Commercial	150–400	Moderate	Honest comparison post including DecodeLabs as an option
8	Data Science internship project ideas for resume 2026	Informational	300–600	Moderate	Curated 8–10 project idea list, each linking deeper
9	Build a real-time chat app project for a data science internship	Informational	30–100	Great	Tutorial walkthrough with GitHub starter repo
10	E-commerce store project ideas for a full stack internship	Informational	50–150	Great	Project brief with React/Node.js stack suggestions
11	TensorFlow projects for beginners internship portfolio	Informational	80–200	Great	Beginner project list with portfolio-presentation angle
12	Pandas vs NumPy — which to learn first for data science	Informational	100–300	Great	Short explainer disambiguating the two libraries

#	Keyword	Intent	Est. Volume (India)	KGR Verdict	Suggested Content Angle
13	React vs Node.js for a full stack developer internship	Commercial	100–250	Moderate	Comparison post tied to internship stack choice
14	Apply for AI/ML internship with stipend for freshers in India	Transactional	150–400	Moderate	Application page, stipend/duration/eligibility upfront
15	Node.js developer internship remote apply 2026	Transactional	80–200	Great	Apply-now page optimized around the exact phrase
16	React internship for final year CSE students apply	Transactional	50–150	Great	Apply-now page targeting final-year eligibility
17	SQL internship projects for data analyst freshers	Informational	100–250	Great	Project list (query optimization, dashboards) + CTA
18	Data Science internship certificate value for placements	Informational	60–150	Great	FAQ/blog post addressing a real student concern
19	Unpaid vs paid data science internship — which to choose	Commercial	40–100	Great	Pros/cons post positioning the paid-track advantage
20	Full Stack Developer internship roadmap for 3rd year students	Informational	150–350	Moderate	Timeline roadmap post (skills by month) + CTA

Legend: Great = KGR < 0.25 (rank fast) | Moderate = KGR 0.25–1.00 (winnable, slower)

7. Calibration: Good vs. Bad Keywords

The training brief itself lists two sample “good” keywords: “Data Scientist salary in India 2026” and “Python vs Java for backend development.” A live SERP check of both tells a more complete story — as written, they are already dominated by large, high-authority publishers (Glassdoor, PayScale, AmbitionBox, upGrad, and Coursera all rank on page one for these exact phrases), which pushes their realistic KGR into ‘Bad’ territory for a new site.

That doesn't make the topics wrong — it makes the phrasing too broad. Both were kept in the final list (rows 1 and 2 of Section 6) but reworded with an audience-specific qualifier — “for freshers with Python and SQL”, “for B.Tech students” — which reclaims a long-tail angle a small site can realistically rank for while preserving the original search intent. This is the practical nuance behind “Specificity is Power”: the long-tail advantage only materializes if the long tail is genuinely underserved, not simply longer.

Applying the calibration rule generally

The Bad (Too Broad)	The Good (High Intent, Long-Tail)
Python	Data Scientist salary in India 2026 for freshers with Python and SQL
Free Internship	Free Python internship certificate for engineering students
What is code?	Remote Data Science internship for freshers with stipend

8. KGR Calculation Reference

Formula: $KGR = \text{allintitle results} \div \text{monthly search volume}$. The worked examples below illustrate the mechanic; plug in live numbers from Keyword Planner and the allintitle: operator to recheck any keyword before publishing.

Keyword	Allintitle Results	Monthly Volume	KGR / Verdict
Free Python internship certificate for engineering students	40	350	0.11 — Great
Remote Data Science internship for freshers with stipend	210	550	0.38 — Moderate
Data Scientist salary in India 2026 (unmodified, for reference)	3,200	14,000	0.23 — Great in theory, Bad in practice*

**Illustrates the exact trap covered in Section 7: the raw KGR arithmetic can clear the 0.25 threshold while the query is still functionally unwinnable, because allintitle counts alone don't capture domain authority. A handful of DA90+ publishers occupying page one outweighs a favorable ratio — always sanity-check the actual top-10 results, not just the numbers.*

9. Content Recommendations

Sequencing suggestion for the first content sprint, ordered by KGR strength and production effort:

- Weeks 1–2: Transactional ‘Great’ keywords (rows 4, 15, 16) — fast to build, direct funnel impact.
- Weeks 3–4: Informational ‘Great’ keywords (rows 5, 6, 9–12, 17, 18) — builds topical authority and internal linking depth.
- Weeks 5–6: Commercial ‘Great’ and ‘Moderate’ keywords (rows 1, 2, 7, 13, 19) — comparison content, slower to rank but high conversion value.
- Ongoing: Moderate-KGR transactional keywords (rows 3, 14) — revisit once the site has built enough authority to compete.

10. Limitations & Verification Notes

Search-intent classification in this report is pattern-based judgment, not proprietary data, and can be trusted as-is. Monthly volume and competition figures, however, are estimates derived from live SERP-crowding checks performed in July 2026 — not pulled directly from Google Keyword Planner or Ahrefs, both of which require an authenticated account outside this project's access. Before this list is locked into a content calendar:

- Confirm exact monthly search volume in Google Keyword Planner (free with a Google Ads account).
- Confirm allintitle counts by running each keyword through Google's allintitle: operator.
- Recalculate KGR with the verified numbers and re-flag any keyword that moves out of the ‘Great’ band.

This satisfies the training brief's own standard — “all projects must be verified for quality” — before the list is handed off for content production.

11. Conclusion

This project reframes “keyword research” from a guessing game into a repeatable process: brainstorm from the target domain, expand with real search modifiers, validate against the live SERP, calculate KGR, and categorize by intent. The resulting 20-keyword shortlist favors specificity over volume on purpose — in a niche already contested by large, well-resourced platforms, long-tail precision is the only lever a smaller site fully controls. The next

milestone, Project 3: Paid Ads, is gated on this verified demand data, in keeping with DecodeLabs' rule of not spending on acquisition until organic intent has been confirmed.

Appendix A — Data Sources

- Google (live search) — SERP crowding checks for each candidate keyword, July 2026.
- Glassdoor India, AmbitionBox, PayScale, upGrad — referenced while assessing competition on salary-related queries.
- Coursera, hackr.io, Talent500 — referenced while assessing competition on language-comparison queries.
- Naukri Campus, Internshala, The Internship — referenced while assessing competition on internship-application queries.
- DecodeLabs Project 2 training brief (Digital Marketing — Industrial Training Kit, Batch 2026) — domain scope, KGR thresholds, and intent framework.